Differential Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solving by separation of variables**

Solve $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$.

Separate: $\frac{dy}{y} = \frac{dx}{x}$. Integrate: $\ln|y| = \ln|x| + C$, so $y = Ax$. Using $y = 6, x = 2$: $A = 3$. Solution: $y = 3x$.

2 **[Foundational]** • **Target: Forming differential equations from a description**

The rate of decrease of a quantity Q is proportional to Q itself. Write a differential equation for $\frac{dQ}{dt}$, using $k > 0$ as the constant of proportionality.

$\frac{dQ}{dt} = -kQ$.

3 **[Foundational]** • **Target: Forming differential equations from a description**

A tank drains so that the rate of decrease of its volume V is proportional to $\sqrt{V}$. Write a differential equation for $\frac{dV}{dt}$, using $k > 0$.

$\frac{dV}{dt} = -k\sqrt{V}$.

4 **[Foundational]** • **Target: Forming differential equations from a description**

A chemical reaction's concentration C decreases at a rate proportional to the square of the concentration. Write a differential equation for $\frac{dC}{dt}$, using $k > 0$.

$\frac{dC}{dt} = -kC^2$.

5 **[Foundational]** • **Target: Solving modeling equations with initial conditions**

A population P satisfies $\frac{dP}{dt} = 0.08P$. If $P = 500$ when $t = 0$, find P when $t = 10$, to the nearest whole number.

$P(t) = 500e^{0.08t}$. $P(10) = 500e^{0.8} = 500(2.225541) \approx 1113$.

6 **[Foundational]** • **Target: Solving modeling equations with initial conditions**

Solve $\frac{dV}{dt} = -k\sqrt{V}$ (from the tank problem above), given $V(0) = V_0$.

Separate: $V^{-1/2}\,dV = -k\,dt$. Integrate: $2\sqrt{V} = -kt + C$. Using $V(0) = V_0$: $C = 2\sqrt{V_0}$. So $\sqrt{V} = \sqrt{V_0} - \frac{kt}{2}$, giving $V(t) = \left(\sqrt{V_0} - \frac{kt}{2}\right)^2$.

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7 [Foundational] • Target: First-order linear differential equations using an integrating factor

Solve $\frac{dy}{dx} + 3y = 12$ using an integrating factor, given $y = 2$ when $x = 0$.

Integrating factor: e^{3x}. $\frac{d}{dx}(ye^{3x}) = 12e^{3x}$. Integrate: $ye^{3x} = 4e^{3x} + C$, so $y = 4 + Ce^{-3x}$. Using $y = 2, x = 0$: $C = -2$. Solution: $y = 4 - 2e^{-3x}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Verifying solutions to differential equations

Verify that $y = 3e^{2x}$ satisfies $\frac{dy}{dx} - 2y = 0$.

$\frac{dy}{dx} = 6e^{2x}$. Substituting: $6e^{2x} - 2(3e^{2x}) = 6e^{2x} - 6e^{2x} = 0$, confirming the equation is satisfied.

9 [Intermediate] • Target: Verifying solutions to differential equations

Verify that $y = x^2 + Ce^{-x}$ is a general solution of $\frac{dy}{dx} + y = x^2 + 2x$.

$\frac{dy}{dx} = 2x - Ce^{-x}$. Substituting: $(2x - Ce^{-x}) + (x^2 + Ce^{-x}) = 2x + x^2$, which matches the right-hand side $x^2 + 2x$ exactly, confirming the equation is satisfied for any constant C.

10 [Intermediate] • Target: General and particular solutions

Find the general solution of $\frac{dy}{dx} = 4x^3 - 6x$, and explain what determines a particular solution from this family.

Integrating directly: $y = x^4 - 3x^2 + C$. A particular solution is obtained by specifying one point that the curve must pass through, which fixes the value of the arbitrary constant C.

11 [Intermediate] • Target: General and particular solutions

A curve satisfies $\frac{dy}{dx} = 6x^2 - 4$ and passes through $(1, -1)$. Find the equation of this particular solution.

$y = 2x^3 - 4x + C$. Using $(1, -1)$: $-1 = 2 - 4 + C \Rightarrow C = 1$. Solution: $y = 2x^3 - 4x + 1$.

12 [Intermediate] • Target: General and particular solutions

A curve satisfies $\frac{dy}{dx} = \frac{1}{x}$ for $x > 0$ and passes through $(1, 4)$. Find the particular solution.

$y = \ln x + C$. Using $(1, 4)$: $4 = 0 + C \Rightarrow C = 4$. Solution: $y = \ln x + 4$.

13 [Intermediate] • Target: Applications: Newton's law of cooling

A cup of tea at $85\,^\circ$C is placed in a room at $22\,^\circ$C. Using $T(t) = T_a + (T_0 - T_a)e^{-kt}$, and given that after 8 minutes the tea has cooled to $65\,^\circ$C, find k, to three decimal places.

$65 = 22 + 63e^{-8k} \Rightarrow e^{-8k} = \frac{43}{63} \approx 0.6825$. Taking natural logs: $-8k \approx -0.3819$, so $k \approx 0.0477$.

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14 [Intermediate] • Target: Applications: Newton's law of cooling

Using the value of k found above, find the temperature of the tea after 20 minutes, to the nearest degree.

$T(20) = 22 + 63e^{-0.0477(20)} = 22 + 63e^{-0.954} \approx 22 + 63(0.3852) \approx 22 + 24.27 \approx 46^\circ$C.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Applications: Newton's law of cooling • Cross-Lesson: 12.01 Poisson Distribution

A metal rod at 200°C cools in air at 25°C, with cooling constant $k = 0.03$ per minute. Find its temperature after 15 minutes, to the nearest degree.

$T(15) = 25 + 175e^{-0.03(15)} = 25 + 175e^{-0.45} \approx 25 + 175(0.6376) \approx 25 + 111.59 \approx 137^\circ$C.

16 [Challenging] • Target: A well-designed word problem: a mixing tank • Cross-Lesson: 13.01 Continuous Random Variables

This question has three parts. A tank contains 300 litres of brine with 10 kg of dissolved salt. Fresh water flows in at 6 litres/minute, and the well-mixed solution flows out at the same rate, keeping the volume constant. Let $S(t)$ be the mass of salt (kg) at time t minutes. (a) Explain why $\frac{dS}{dt} = -\frac{S}{50}$. (b) Solve this differential equation for $S(t)$, given $S(0) = 10$. (c) Find how long it takes for the salt content to drop to 2 kg, to the nearest minute.

(a) Since the incoming water carries no salt, salt only leaves, at a rate equal to concentration times outflow: $\frac{S}{300} \times 6 = \frac{S}{50}$ kg/min, so $\frac{dS}{dt} = -\frac{S}{50}$. (b) Separate: $\frac{dS}{S} = -\frac{dt}{50}$. Integrate: $\ln S = -\frac{t}{50} + C$, so $S = Ae^{-t/50}$. Using $S(0) = 10$: $A = 10$. Solution: $S(t) = 10e^{-t/50}$. (c) $2 = 10e^{-t/50} \Rightarrow e^{-t/50} = 0.2 \Rightarrow -\frac{t}{50} = \ln 0.2 \approx -1.6094 \Rightarrow t \approx 80$ minutes.

17 [Challenging] • Target: Applications: Newton's law of cooling • Cross-Lesson: 14.01 Sampling & Estimation

The temperature θ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where t is time in minutes and k is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of t when $\theta = 30$.

Separate variables and integrate: $\int \frac{1}{\theta - 20}\, d\theta = -\int k\, dt \Rightarrow \ln(\theta - 20) = -kt + C$ Apply initial condition $t = 0, \theta = 80$: $\ln(60) = C \Rightarrow \ln\left(\frac{\theta - 20}{60}\right) = -kt \Rightarrow \theta - 20 = 60e^{-kt}$ Apply condition $t = 10, \theta = 50$: $30 = 60e^{-10k} \Rightarrow e^{-10k} = \frac{1}{2} \Rightarrow k = \frac{\ln 2}{10} \approx 0.06931$ Set $\theta = 30$.

18 [Challenging] • Target: A well-designed word problem: a mixing tank • Cross-Lesson: 15.01 Hypothesis Testing

Solve the first-order linear differential equation $\frac{dy}{dx} + \frac{2}{x}y = 4x$ for $x > 0$, given the initial condition $y(1) = 2$.

1. Identify $P(x) = \frac{2}{x}$ and $Q(x) = 4x$. 2. Calculate the integrating factor $\mu(x)$: $\mu(x) = \exp\left(\int \frac{2}{x}\, dx\right) = e^{2\ln x} = e^{\ln(x^2)} = x^2$. 3. Multiply through by x^2: $x^2\frac{dy}{dx} + 2xy = 4x^3 \Rightarrow \frac{d}{dx}[x^2 y] = 4x^3$. 4. Integrate both sides: $x^2 y = \int 4x^3\, dx = x^4 + C$. 5. Apply $y(1) = 2$: $(1)^2(2) = 1^4 + C \Rightarrow 2 = 1 + C \Rightarrow C = 1$. 6. Explicit solution: $y = \frac{x^4 + 1}{x^2} = x^2 + \frac{1}{x^2}$.

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19 [Challenging] • Target: Solving by separation of variables • Cross-Lesson: 2.01 Exponentials & Logarithms

Solve the second-order homogeneous linear differential equation $y'' - 5y' + 6y = 0$ satisfying initial conditions $y(0) = 1$ and $y'(0) = 3$.

1. Write the characteristic equation: $m^2 - 5m + 6 = 0 \Rightarrow (m-2)(m-3) = 0 \Rightarrow m_1 = 2,\ m_2 = 3.$ 2. General solution: $y(x) = Ae^{2x} + Be^{3x}.$ 3. Derivative: $y'(x) = 2Ae^{2x} + 3Be^{3x}.$ 4. Apply initial conditions: $y(0) = 1 \Rightarrow A + B = 1.$ $y'(0) = 3 \Rightarrow 2A + 3B = 3.$ Multiply first equation by 2: $2A + 2B = 2.$ Subtract from second: $B = 1.$ Then $A = 1 - 1 = 0.$ Particular Solution: $y(x) = e^{3x}.$

20 [Challenging] • Target: Forming differential equations from a description • Cross-Lesson: 3.01 Advanced Trigonometry

[Differential Equations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solving by separation of variables

First-order linear ODE solution structure: general solution = homogeneous solution +

a. **particular solution ✓**

b. another homogeneous solution

c. the initial condition

d. zero

Applying Differential Equations principles, $y = y_h + y_p$. Therefore, the verified correct option is particular solution.

22 [Foundational] • Target: Forming differential equations from a description

A radioactive substance has half-life 2 years and initial mass 16g. Find the mass remaining after 4 years.

a. **4 ✓**

b. 16

c. 8

d. 5

After 2 half-life period(s) (4 years), mass $= 16/2^2 = 4$g.

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23 [Foundational] • Target: Solving modeling equations with initial conditions

Solve $y' - \frac{y}{x} = x^2$ (linear first-order):

a. $y = \frac{1}{2}x^3 + Cx$ ✓

b. $y = \frac{1}{3}x^3 + C$

c. $y = x^3 + \frac{C}{x}$

d. $y = x^3 + Cx$

The integrating factor is $\mu(x) = e^{-\int \frac{1}{x}\,dx} = \frac{1}{x}$. Then $\left(\frac{y}{x}\right)' = x \Rightarrow \frac{y}{x} = \frac{x^2}{2} + C \Rightarrow y = \frac{1}{2}x^3 + Cx$.

24 [Foundational] • Target: First-order linear differential equations using an integrating factor

Coffee at 70°C is placed in a 22°C room and cools to 45°C in 5 min (Newton's law of cooling). How long (min) until it reaches 35°C?

a. 7

b. $\frac{10\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$

c. $\frac{5\ln(2)}{\ln\left(\frac{14}{9}\right)}$

d. $\frac{5\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$ ✓

$T - 22 = 48e^{-kt}$ with $k = \frac{\ln\left(\frac{48}{23}\right)}{5}$. Solve $48e^{-kt} = 13$: $t = \frac{5\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$.

25 [Foundational] • Target: Further integrating factor examples

Newton's law of cooling: $\frac{dT}{dt} = k(T - T_s)$, $k < 0$. Steady state:

a. $T = 0$

b. $T = kT_s$

c. $T = T_s$ ✓

d. $T = T_0$

Applying Differential Equations principles, $T \to T_s$ as $t \to \infty$. Therefore, the verified correct option is $T = T_s$.

26 [Foundational] • Target: Verifying solutions to differential equations

A first-order linear ODE has the form:

a. $y' + P(x)y = Q(x)$ ✓

b. $y' = P(x)y^2$

c. $y'y = Q(x)$

d. $y'' = P(x)$

Applying Differential Equations principles, Standard form: $y' + P(x)y = Q(x)$. Therefore, the verified correct option is $y' + P(x)y = Q(x)$.

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27 [Foundational] • Target: General and particular solutions

For the linear ODE $y' + 2y = 4$: integrating factor $\mu =$

a. e^2

b. 2

c. e^{-2x}

d. e^{2x} ✓

Applying Differential Equations principles, $\mu = e^{\int 2\,dx} = e^{2x}$. Therefore, the verified correct option is e^{2x}.

28 [Foundational] • Target: Applications: Newton's law of cooling

Linear homogeneous 2nd-order ODE: if y_1, y_2 are solutions, so is:

a. y_1^2

b. $y_1 y_2$

c. y_1'

d. $y_1 + y_2$ ✓

Applying Differential Equations principles, Superposition principle. Therefore, the verified correct option is $y_1 + y_2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a mixing tank

Which differential equation correctly models the statement: "The rate of increase of a population P is proportional to P itself."?

a. $\frac{dP}{dt} = kP$ ✓

b. $\frac{dT}{dt} = -k(T - T_e)$

c. $\frac{dm}{dt} = -km$

d. $\frac{dN}{dt} = kN(S - N)$

The correct model is $\frac{dP}{dt} = kP$, where $k > 0$ is a constant of proportionality.

30 [Intermediate] • Target: Solving by separation of variables

Find the general solution of $\frac{dy}{dx} = 2x^2$.

a. $y = \frac{2x^3}{3} + C$ ✓

b. $y = \frac{2x^3}{3}$

c. $y = 6x^3 + C$

d. $y = 2x^2 + C$

Direct integration: $y = \int 2x^2\,dx = \frac{2x^3}{3} + C$.

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31 [Intermediate] • Target: Forming differential equations from a description

A quantity satisfies $\frac{dy}{dt} = -y$ with $y(0) = 9$. Find y when $t = \ln(3)/1$.

a. 4

b. 27

c. **3** ✓

d. 9

$y = y_0 e^{-t}$. At the given t, $e^{-t} = 1/3$, so $y = 3$.

32 [Intermediate] • Target: Solving modeling equations with initial conditions

Evaluate $\int_0^1 xe^x\,dx$ using integration by parts.

a. -1

b. **1** ✓

c. $1 + e^2$

d. 2

$\int_0^1 xe^x\,dx = 1$.

33 [Intermediate] • Target: First-order linear differential equations using an integrating factor

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

a. $e^{-1} + 1$

b. $2e^{-1} + 1$ ✓

c. $e^{-1} - 1$

d. $2 - e^{-1}$

The given differential equation for the y-component of the velocity is $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0$, which factors as $(r-1)^2 = 0$. Thus, we have a repeated root $r = 1$. The general solution for $y(t)$ is $y(t) = c_1 e^t + c_2 te^t$. Using the initial conditions $y(0) = 2$ and $y'(0) = 1$: $y(0) = c_1 e^0 + c_2(0)e^0 = c_1 = 2$. $y'(t) = 2e^t + 2te^t + c_2 e^t + c_2 te^t = (2 + c_2)e^t + (2 + c_2)te^t$.
$y'(0) = (2 + c_2)e^0 + (2 + c_2)(0)e^0 = 2 + c_2 = 1$, so $c_2 = -1$. Thus.

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34 [Intermediate] • Target: Further integrating factor examples

Newton's law of cooling: $T(t) = T_e + (T_0 - T_e)e^{-kt}$. If room temp $T_e = 20$, initial $T_0 = 80$, $k = 0.1$, $T(10) =$

a. $20 + 60$

b. $60e^{-1}$

c. $80e^{-1}$

d. $20 + 60e^{-1} \approx 42.1$ ✓

Applying Transcendental Functions principles, $T(10) = 20 + 60e^{-1} \approx 42.1°\text{C}$. Therefore, the verified correct option is $20 + 60e^{-1} \approx 42.1$.

35 [Intermediate] • Target: Verifying solutions to differential equations

A second-order ODE with repeated root $r = r_1$ has general solution:

a. $y = C_1e^{r_1x} + C_2e^{-r_1x}$

b. $y = C_1e^{r_1x}$

c. $y = (C_1 + C_2)e^{2r_1x}$

d. $y = (C_1 + C_2x)e^{r_1x}$ ✓

Applying Differential Equations principles, Repeated root: multiply by x. Therefore, the verified correct option is $y = (C_1 + C_2x)e^{r_1x}$.

36 [Intermediate] • Target: General and particular solutions

A 200-L tank holds brine with 20 kg of salt. Brine with $\frac{1}{2}$ kg/L enters at 10 L/min and the well-mixed solution leaves at the same rate. How much salt (kg) is in the tank after 20 min?

a. $100 - \frac{80}{e}$ ✓

b. $100 - \frac{100}{e}$

c. 120

d. $\frac{20}{e} + 100$

$A' = 5 - \frac{10}{200}A$, a linear equation with equilibrium 100. So $A = 100 - 80e^{-\frac{t}{20}}$ and $A(20) = 100 - \frac{80}{e}$.

37 [Intermediate] • Target: Applications: Newton's law of cooling

Variation of parameters for $y'' + y = \sec x$: general solution involves $\ln|\sec x + \tan x|$. This appears because the particular integral of $\int \sin x \sec x\, dx =$

a. $\tan x$

b. $\ln|\cos x|$

c. $-\ln|\cos x|$ ✓

d. $-\cos x$

Applying Differential Equations principles, $\int \tan x\, dx = -\ln|\cos x|$. Therefore, the verified correct option is $-\ln|\cos x|$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: A well-designed word problem: a mixing tank** • **Cross-Lesson: 15.01 Hypothesis Testing**
Integrating factor for $y' + P(x)y = Q(x)$:

a. $e^{P(x)}$

b. $e^{\int P\,dx}$ ✓

c. $P(x)$

d. $1/P(x)$

Applying Differential Equations principles, $\mu = e^{\int P\,dx}$. Therefore, the verified correct option is $e^{\int P\,dx}$.

39 **[Challenging]** • **Target: Solving by separation of variables** • **Cross-Lesson: 2.01 Exponentials & Logarithms**
Initial value problem: $y' = y$, $y(0) = 2$. Solution:

a. $y = e^{2x}$

b. $y = 2e^{x}$ ✓

c. $y = 2x + 2$

d. $y = xe^{x}$

Applying Differential Equations principles, $y = Ce^{x}$; $y(0) = C = 2$. Therefore, the verified correct option is $y = 2e^{x}$.

40 **[Challenging]** • **Target: Forming differential equations from a description** • **Cross-Lesson: 3.01 Advanced Trigonometry**
Second-order ODE with complex roots $r = \alpha \pm \beta i$: general solution includes:

a. $e^{i\beta x}$

b. $C_1 e^{\alpha x} + C_2 e^{\beta x}$

c. $(C_1 + C_2 x)e^{(\alpha+\beta)x}$

d. $e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$ ✓

Euler: $e^{\alpha x}$ envelope times sinusoids.

41 **[Challenging]** • **Target: Solving modeling equations with initial conditions** • **Cross-Lesson: 4.01 Differentiation**
Fundamental theorem for linear ODEs: the general solution of $Ly = g$ is:

a. y_p only

b. $y_h + y_p$ ✓

c. y_h only

d. any solution of $Ly = 0$

General = homogeneous + particular.

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42 [Challenging] • Target: First-order linear differential equations using an integrating factor • Cross-Lesson: 5.01 Integration

The first, fourth, and sixth terms of an arithmetic progression (u_n) are equal to the first, second, and third terms, respectively, of an infinite geometric progression (v_n) with common ratio $re1$. Given that the sum to infinity of the geometric progression is 27, determine the maximum value of $S_n = \sum_{k=1}^{n} u_k$, the sum of the first n terms of the arithmetic progression.

a. 44

b. 45 ✓

c. 50

d. 54

Let the arithmetic progression (u_n) have first term a and common difference d. Then: $u_1 = a, \quad u_4 = a + 3d, \quad u_6 = a + 5d$. Let the geometric progression (v_n) have first term v_1 and common ratio r. Since $u_1 = v_1$, $u_4 = v_2$, and $u_6 = v_3$, we have: $v_1 = a, \quad v_2 = ar, \quad v_3 = ar^2$. Equating terms: 1) $a + 3d = ar \Rightarrow 3d = a(r-1) \Rightarrow d = \frac{a(r-1)}{3}$. 2) $a + 5d = ar^2$. Substitute d into the second equation.

43 [Challenging] • Target: Further integrating factor examples • Cross-Lesson: 6.01 Sequences & Series

The first three terms of a geometric progression are given by $u_1 = 3k - 2, \quad u_2 = k + 2, \quad u_3 = k - 2,$ where k is a real constant. Given that the infinite geometric series $\sum_{n=1}^{\infty} u_n$ converges, find the sum to infinity, S_∞.

a. 32 ✓

b. -1

c. $\frac{32}{3}$

d. 16

Step 1: Set up the common ratio property of a geometric sequence. Since the terms form a geometric progression, the ratio between consecutive terms must be equal: $\frac{u_2}{u_1} = \frac{u_3}{u_2} \Rightarrow u_2^2 = u_1 u_3$ Step 2: Substitute the expressions in terms of k: $(k+2)^2 = (3k-2)(k-2)$ $k^2 + 4k + 4 = 3k^2 - 8k + 4$ Step 3: Solve the quadratic equation for k: $2k^2 - 12k = 0 \Rightarrow 2k(k-6) = 0$ Thus, $k = 0$ or $k = 6$. Step 4: Apply the convergence criterion $|r| < 1$: - If $k = 0$.

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44 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 7.01 Vectors

The continuous random variable X has probability density function given by $f(x) = \{k(4 - x^2), 0 \le x \le 2, ; 0, \text{otherwise},$ where k is a constant. Find the value of k.

a. $\frac{3}{16}$ ✓

b. $\frac{1}{4}$

c. $\frac{3}{8}$

d. $\frac{16}{3}$

For $f(x)$ to be a valid probability density function, the total area under the curve must equal 1: $\int_{-\infty}^{\infty} f(x)\,dx = 1$ Substitute the non-zero definition of $f(x)$: $\int_0^2 k(4 - x^2)\,dx = 1$ Factor out the constant k and integrate term by term: $k\left[4x - \frac{x^3}{3}\right]_0^2 = 1$ Evaluate at the upper and lower limits: $k\left(\left(4(2) - \frac{2^3}{3}\right) - 0\right) = 1$ $k\left(8 - \frac{8}{3}\right) = 1$.

45 [Challenging] • Target: General and particular solutions • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

The first three terms of a geometric progression are 32, 24, and 18. Find the sum to infinity, S_∞, of the progression.

a. 128 ✓

b. 96

c. 74

d. 64

The first term of the geometric progression is $a = 32$. The common ratio r is found by dividing any term by the preceding term: $r = \frac{24}{32} = \frac{3}{4}$ Since $|r| = \left|\frac{3}{4}\right| < 1$, the geometric progression converges and the sum to infinity exists. The sum to infinity is given by: $S_\infty = \frac{a}{1 - r}$ Substituting $a = 32$ and $r = \frac{3}{4}$: $S_\infty = \frac{32}{1 - \frac{3}{4}} = \frac{32}{\frac{1}{4}} = 32 \times 4 = 128$ Common misconceptions: - Choice B (96) arises fro.

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